Patient Cirrhosis Analytics and Predictive Modeling

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Problem Statement:

Health organizations face major challenges regarding liver diseases, especially liver cirrhosis that it may lead to liver failure making the patients need to a liver transplant or in the worst case scenario death. In light of rapidly significant developments in Saudi Arabia that align with the Saudi 2030 vision, one of the major objectives of the vision is to enhance the healthcare sector's efficiency by integrating technology. Moreover, this project aims to implement a comprehensive analysis, including a machine learning model, such as a decision tree, to gain insights into cirrhosis by understanding key factors in the dataset, including medication, clinical indicators, and laboratory tests. The goal is to identify high-risk patients with liver cirrhosis and identify their patterns. This project reflects a positive impact on society by reducing costs in the healthcare sector and supporting the Vision 2030 goal in reaching a healthier community.

Objectives:

The Primary objectives of this project are as follows:

- Understanding the patient's characteristics in the dataset, such as disease stage, blood tests, and other variables.
- Preprocessing the data before building the model, such as handling missing values, noisy data, and outliers.
- Using clustering, statistical methods, and visualization to understand the relationship between data observations and determine interesting patterns.
- Selection of the best Decision Tree model that predicts a patient's status based on Evaluation metrics and performance measures.

Hypotheses:

- **H1**: Cirrhosis stage is one of the strongest predictors of patient status.
- **H2**: Class imbalance is expected to affect the performance of the decision tree model.
- **H3**: Patients who receive the drug are more likely to have censored status.
- **H4**: Higher cirrhosis stage is associated with a higher risk of death.

Project Questions:

- What are the main patterns in patients across status groups?
- Which features are the most useful in distinguishing between statuses?
- How well can the model predict status despite class imbalance?
- How do treatment and cirrhosis contribute in explaining variety of patient status?

Data Source:

UCI Cirrhosis dataset, was selected because it contains well-organized data with 17 mixed features, including missing values. The dataset is crucial for predicting patient cirrhosis status.

Features in the dataset include:

- Demographic features:

Age, Sex

- Laboratory tests:

Bilirubin, Cholesterol, Albumin, Copper, Alk_Phos, SGOT, Tryglicerides, Platelets, and Prothrombin

- Medical history:

Ascites, Hepatomegaly, Spiders, Edema, and Stage

- Medicine/treatment variable:

Drug

The dataset can be accessed here: UCI Cirrhosis Dataset

Data Description:

ID: ID of the patient

N_DAYS: Number of days between status change or completeness of the study.

STATUS: status of the patient C (censored), CL (censored due to liver transplant), or D (death)

DRUG: Type of drug

AGE: Patient age

SEX: Male of Female

ASCITES: Presence of ascites N (No) or Y (Yes)

HEPATOMEGALY: Presence of hepatomegaly N (No) or Y (Yes)

SPIDERS: Presence of spiders N (No) or Y (Yes)

EDEMA: presence of edema N, S edema present without diuretics, or edema resolved by diuretics, or Y edema despite diuretic therapy

BILIRUBIN: Bilirubin

CHOLESTEROL: Cholesterol

ALBUMIN: Albumin

COPPER: Copper

ALK_PHOS: Alk_Phos

SGOT: SGOT

TRYGLICERIDES: Tryglicerides

PLATELETS: Platelets

PROTHROMBIN: Prothrombin

STAGE: Stage of disease progressing

The collected data contains 418 rows with 20 columns. The columns contain data of different types, including binary, nominal, ordinal, and numeric. SAS Discover Information Assets was used to get information about data types The Data type descriptions in the table below:

FEATURES	DATA TYPE
ID	ID
N_DAYS	Interval
STATUS	Nominal
DRUG	Nominal
AGE	Interval
SEX	Binary
ASCITES	Binary
HEPATOMEGALY	Binary
SPIDERS	Binary
EDEMA	Ordinal
BILIRUBIN	Interval
CHOLESTEROL	Interval
ALBUMIN	Interval
COPPER	Interval
ALK_PHOS	Interval
SGOT	Interval
TRYGLICERIDES	Interval
PLATELETS	Interval
PROTHROMBIN	Interval
STAGE	Ordinal

After exploring datatypes, the next step is to get statistics of both categorical and numeric data. Categorical data statistics involve **different values, mode, and number of missing values**, while on the other hand, numeric data statistics **involve range, median, mean, Standard deviation, and number of missing values.**

Statistics for both categorical and numeric data are listed in the tables below. After data cleaning, these tables will be compared with the statistics of the categorical and numeric data to gain insights into how data cleaning affects the data statistics.

Categorical Variables Table Before Preprocessing

FEATURES	VALUES	MODE	MISSING VALUES
STATUS	C, CL, or D	C	0
DRUG	D- penicillamine or Placebo	D-penicillamine	106
SEX	M, F	F	0
ASCITES	Y, N	N	106
HEPATOMEGALY	Y, N	Y	106
SPIDERS	Y, N	N	106
EDEMA	Y, S, or N	N	0
STAGE	1, 2, 3, or 4	3	6

Numerical Variables Table Before Preprocessing

Features	Range	Median	MEAN	Standard deviation	Missing values
N_Days	(41 – 4795)	1730	1917.782	1104.673	0
Age	(9598 – 28650)	18628	18533.35	3815.845	0
Bilirubin	(0.3 – 28)	1.4	3.220813	4.407506	0
Cholesterol	(120 – 1775)	309.5	369.5106	231.9445	134
ALBUMIN	(1.96 – 4.64)	3.53	3.49744	0.424972	0
COPPER	(4 – 588)	73	97.64839	85.61392	108
ALK_PHOS	(289 – 13862.4)	1259	1982.656	2140.389	106
SGOT	(26.35 – 457.25)	114.7	122.5563	56.69952	106
TRYGLICERIDES	(33 – 598)	108	124.7021	65.14864	136
Platelets	(62 – 721)	251	257.0246	98.32558	11
Prothrombin	(9 – 18)	10.6	10.73173	1.022	2

Data Preprocessing

As seen in the pre-cleaning statistical description of the numeric and categorical data columns, there are 106 missing values across multiple columns for the same data records. These missing values are expected; the data source explicitly stated that 106 patients are not in the clinical trial; however, they agreed to record their basic metrics and undergo survival tracking. There are other missing values in other columns. Moreover, our approach to data preprocessing is to fill missing values by taking the median of each missing value column per patient stage. This approach was chosen to align with the data specific feature. The data is for a specific disease with multiple stages of progression. Using an appropriate statistical measure for each column within each disease stage is expected to reduce imputation bias. Data preprocessing was implemented using SAS Studio Develop Code and Flows, Figure below. The flow started by importing the dataset via an import file node, then provided to a Python code node. After that, it moved to a table node to check whether the data cleaning process was completed.



Python code for data preprocessing fills missing numeric values with the median of each column per stage. The median was used to reduce the effect of outliers, as medical data varies from patient to patient. On the other hand, the mode is used to fill missing categorical data, except for the Drug column, where the 106 missing data were filled by creating a new type called No drug for the 106 that are not in the clinical trial. The age column is defined as the age of the patient in days. This column was divided by 365.25 to calculate the patient's age in years. Furthermore, the next step is to recalculate categorical and numeric statistics to compare them with the column statistics in tables before data preprocessing.

Tables below show statistics for categorical and numerical features after data preprocessing.

FEATURES	VALUES	MODE	MISSING VALUES
STATUS	C, CL, or D	C	0
DRUG	D-penicillamine, Placebo, No Drug	D-penicillamine	0
SEX	M, F	F	0
ASCITES	Y, N	N	0
HEPATOMEGALY	Y, N	N	0
SPIDERS	Y, N	N	0
EDEMA	Y, S, or N	N	0
STAGE	1, 2, 3, or 4	3	0

Features	Range	Median	MEAN	Standard deviation	Missing values
N_Days	(41 – 4795)	1730	1917.782	1104.673	0
Age	(26.27789 – 78.43943)	51.00068	50.74155	10.44721	0
Bilirubin	(0.3 – 28)	1.4	3.220813	4.407506	0
Cholesterol	(120 – 1775)	296	316.634	112.022	0
ALBUMIN	(1.96 – 4.64)	3.53	3.49744	0.424972	0
COPPER	(4 – 588)	67.5	91.36722	75.0959	0
ALK_PHOS	(289 – 13862.4)	1257.5	1800.852	1876.389	0
SGOT	(26.35 – 457.25)	112.375	120.0116	49.53625	0
TRYGLICERIDES	(33 – 598)	106	119.3182	54.30866	0
Platelets	(62 – 721)	252	257.128	97.09381	0
Prothrombin	(9 – 18)	10.6	10.73098	1.020029	0

Outliers handling

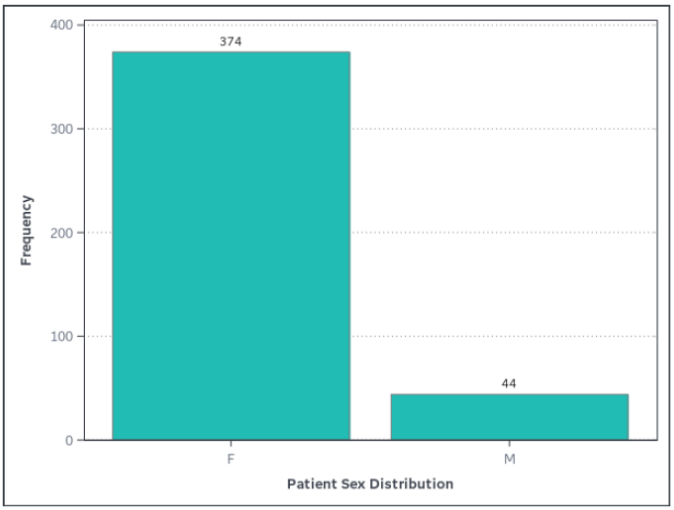
Specifically, some of the features showed extreme values, especially in a medical context, where laboratory measurements vary between patients these values weren't removed due to the fact that these values represent real medical cases.

Exploratory Data Analysis

Exploratory Data Analysis is one of the key tools to get valuable insights about the data. It shows different perspectives on data, including how the data is distributed, data frequency per class, and the Interquartile range.

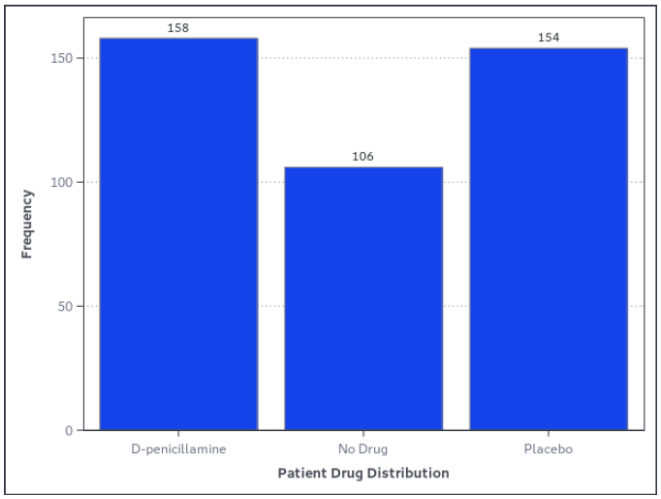
1. Patient Sex Distribution

Bar chart of patient sex distribution shows that females are the most represented in the data; most patients recorded are females. This might be due to the fact that the data is for PBC cirrhosis, which affects females more than males.



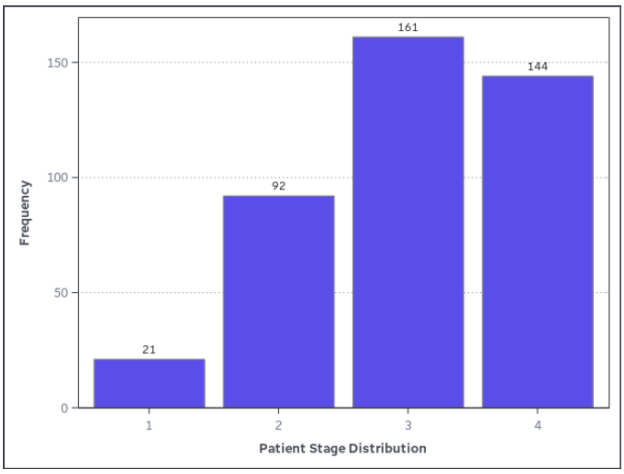
2. Patient Drug Distribution

Patient drug distribution bar chart shows that drug attributes have 3 distinct values: D-penicillamine, Placebo, and no drug. D-penicillamine and placebo frequencies are 158 and 154, respectively, but the No drug frequency is 106, referring to the 106 patients who are not in the study but whose metrics were recorded.



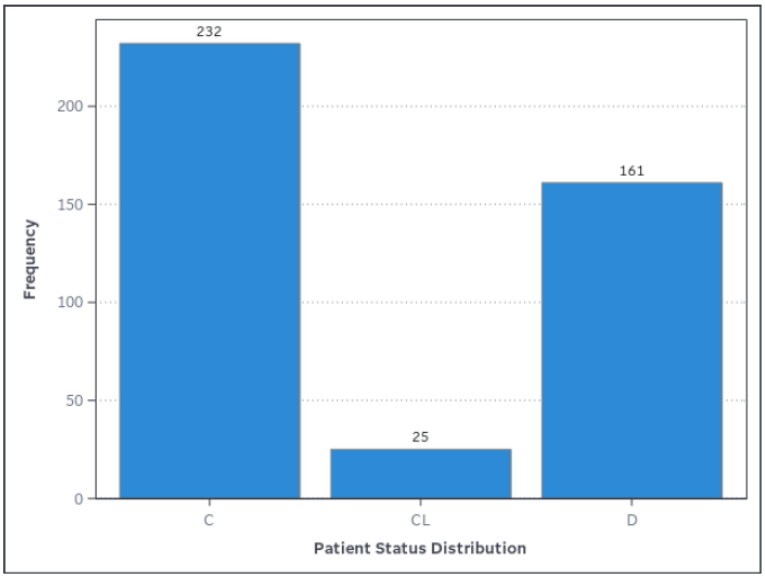
3. Patient Stage Distribution

Patient stage distribution bar chart shows that the two most frequent stages in the data are stage 3 then stage 4.



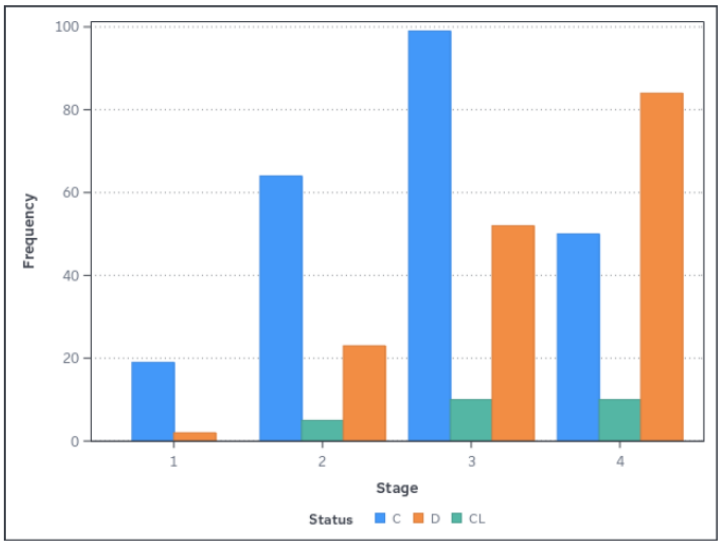
4. Patient Status Distribution

Bar chart of Patient status distribution shows that the status attribute is imbalanced. Censored patients are 232, the Death patients are 161, but for patients that were censored due to liver transplant, there are 25. This might affect patient status prediction due to class imbalance.



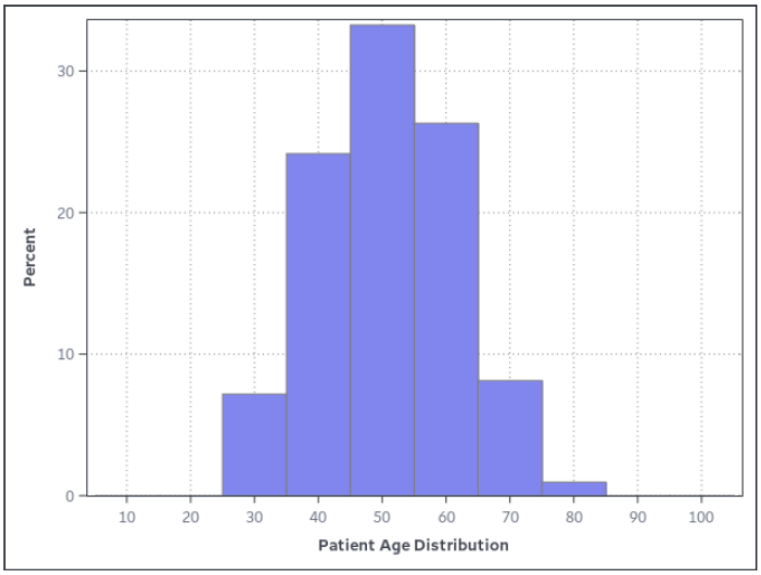
5. Patient Stage by Status

Stage by Status grouped bar chart implies that stage 4 patients are more likely to have D status, this shows how the severity of disease affects the status outcome.



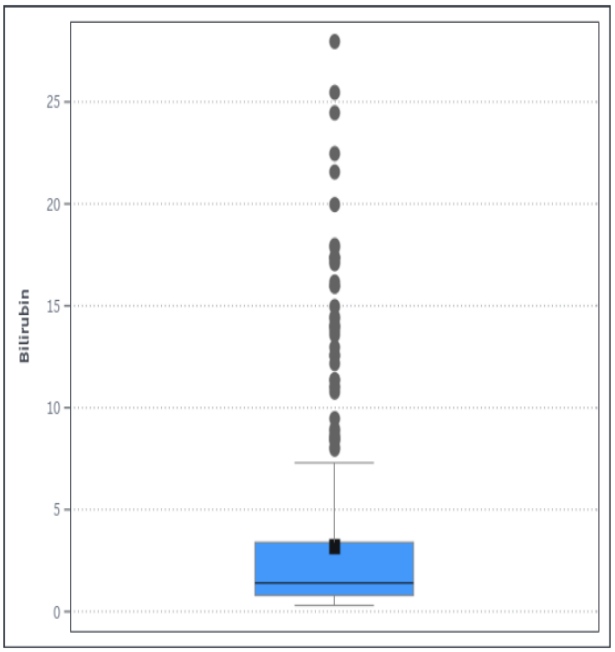
6. Patient Age Distribution

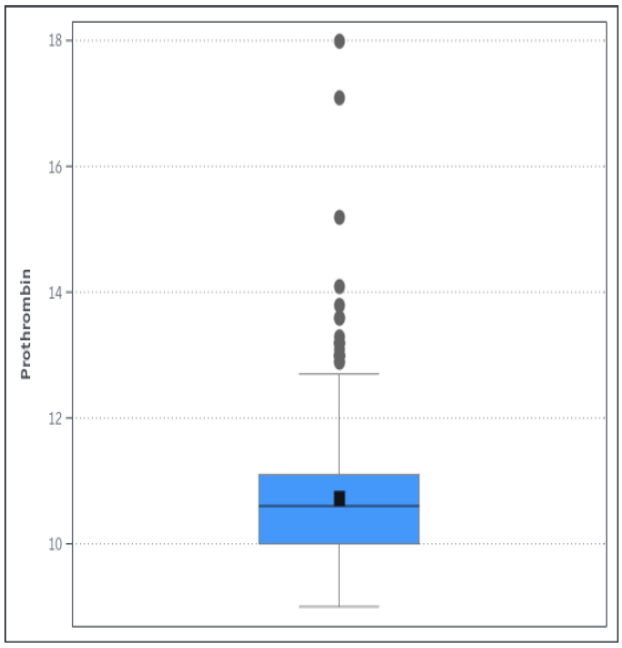
Patient age distribution histogram shows that most patients are between 35 and 65 years old.



7. Patient Bilirubin and Prothrombin Distribution

Box plots for both bilirubin and prothrombin show the interquartile ranges, the mean, and the data points outside the 25-75% range.





Correlation

Pearson Correlation

Pearson correlation measures the strength and direction of the linear relationship between numeric variables. 0-0.4 indicates a weak relationship, 0.41-0.6 a moderate relationship, and 0.61-1 a strong relationship. - sign means inverse direction, + means direct relationship. However, our data does not show any strong linear relationship, but this doesn't mean there is no relationship at all, it only means our numerical attributes are not strongly linearly correlated.

11 Variables: Age N_Days Bilirubin Cholesterol Albumin Copper Alk_Phos SGOT Tryglicerides Platelets Prothrombin

Pearson Correlation Coefficients, N = 418

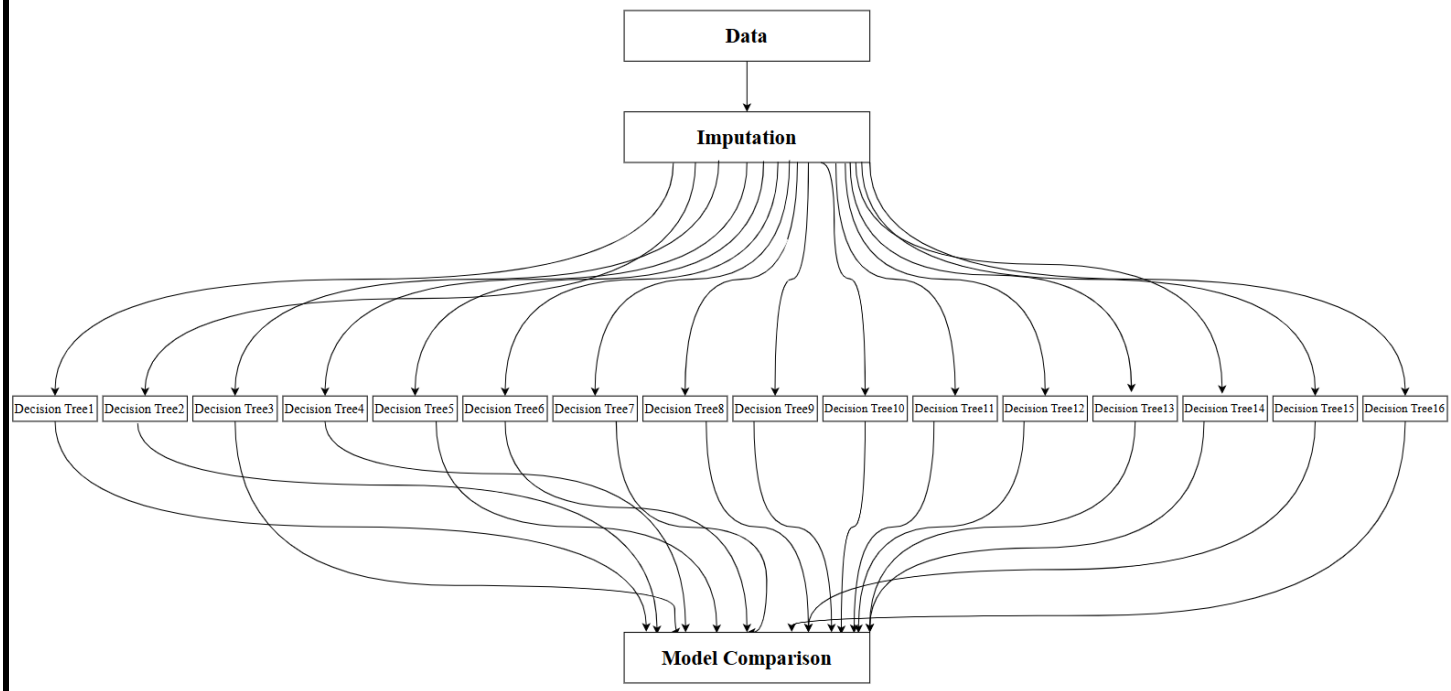
	Age	N_Days	Bilirubin	Cholesterol	Albumin	Copper	Alk_Phos	SGOT	Tryglicerides	Platelets	Prothrombin
Age	1.00000	-0.12593	0.00236	-0.10666	-0.18235	0.04976	-0.05570	-0.12854	0.00629	-0.15014	0.11419
N_Days	-0.12593	1.00000	-0.40395	-0.04545	0.43083	-0.31658	0.14701	-0.19388	-0.11861	0.15078	-0.11282
Bilirubin	0.00236	-0.40395	1.00000	0.20363	-0.31418	0.40452	0.10612	0.39304	0.36635	-0.01279	0.31471
Cholesterol	-0.10666	-0.04545	0.20363	1.00000	-0.00577	0.13544	0.13552	0.20148	0.28004	0.13888	-0.11663
Albumin	-0.18235	0.43083	-0.31418	-0.00577	1.00000	-0.22116	-0.07428	-0.18778	-0.07292	0.15488	-0.19842
Copper	0.04976	-0.31658	0.40452	0.13544	-0.22116	1.00000	0.20906	0.30563	0.28830	-0.05560	0.18747
Alk_Phos	-0.05570	0.14701	0.10612	0.13552	-0.07428	0.20906	1.00000	0.12916	0.18573	0.12737	0.07589
SGOT	-0.12854	-0.19388	0.39304	0.20148	-0.18778	0.30563	0.12916	1.00000	0.13869	-0.10920	0.10101
Tryglicerides	0.00629	-0.11861	0.36635	0.28004	-0.07292	0.28830	0.18573	0.13869	1.00000	0.07972	0.00711
Platelets	-0.15014	0.15078	-0.01279	0.13888	0.15488	-0.05560	0.12737	-0.10920	0.07972	1.00000	-0.15165
Prothrombin	0.11419	-0.11282	0.31471	-0.11663	-0.19842	0.18747	0.07589	0.10101	0.00711	-0.15165	1.00000

Modeling and Final Analysis

Modeling and Final Analysis represents the final analysis of the cirrhosis dataset. It fulfills the data mining life cycle to understand the patient characteristics, identify patterns, and build classification models. Decision Trees and K-means clustering were used to support the Modeling and final analysis, and to help select the 'best' model to achieve accurate prediction, and to provide a clear understanding of how patients can be grouped or classified based on their medical attributes.

Task 1: Decision Tree Modeling

Process Flow Diagram



1.1 Evaluation Approach

Measure	Description	Definition of Value Function	Weight	Threshold
Accuracy	Measures the percentage of test records that are correctly classified by the model.	Accuracy = 1 - Error rate	0.60	> 0.70
Simplicity	Measures how easy the decision tree is to understand. A simpler tree has fewer rules or leaves.	Based on number of leaves. Best range is 5 – 8 leaves.	0.05	None
Lift	Measures how much better the model performs compared with random guessing for the target event.	Lift Score = Tree Lift / Best Possible Lift at the 2nd decile.	0.20	> 0
Stability	Measures whether the model gives steady performance when applied to different data.	min(Train Accuracy, Test Accuracy) / max(Train Accuracy, Test Accuracy)	0.15	> 0.90

1.2 Summary of Results

DT Label/ Description	Performance Measures								
	Accuracy		Simplicity		Lift		Stability		Overall Score
	% Error	Score	No. of Leaves	Score	Value	Score	Score	Justification	
Decision Tree1	0.264	0.736	12	0.200	2.34	0.89	0.98	The closer score to 1 means its more stable	0.777
Decision Tree2	0.384	0.616	2	0	1.45	0.55	0.93	The closer score to 1 means its more stable	0.620
Decision Tree3	0.264	0.736	14	0	2.34	0.89	0.98	The closer score to 1 means its more stable	0.766
Decision Tree4	0.384	0.616	2	0	1.45	0.55	0.93	The closer score to 1 means its more stable	0.620
Decision Tree5	0.288	0.712	2	0	1.8	0.68	0.96	The closer score to 1 means its more stable	0.707
Decision Tree6	0.288	0.712	4	0.667	2.24	0.85	0.96	The closer score to 1 means its more stable	0.775
Decision Tree7	0.288	0.712	2	0	1.8	0.68	0.96	The closer score to 1 means its more stable	0.707
Decision Tree8	0.24	0.760	7	1	2.30	0.87	0.96	The closer score to 1 means its more stable	0.827
Decision Tree9	0.24	0.760	4	0.667	2.14	0.81	0.95	The closer score to 1 means its more stable	0.796
Decision Tree10	0.24	0.760	10	0.6	2.24	0.85	0.94	The closer score to 1 means its more stable	0.798
Decision Tree11	0.288	0.712	2	0	1.8	0.68	0.96	The closer score to 1 means its more stable	0.707
Decision Tree12	0.24	0.760	7	1	2.44	0.93	0.94	The closer score to 1 means its more stable	0.833
Decision Tree13	0.256	0.744	6	1	1.88	0.71	0.96	The closer score to 1 means its more stable	0.785
Decision Tree14	0.256	0.744	6	1	1.88	0.71	0.96	The closer score to 1 means its more stable	0.785
Decision Tree15	0.224	0.776	12	0.2	2.04	0.77	0.97	The closer score to 1 means its more stable	0.777
Decision Tree16	0.224	0.776	12	0.2	2.04	0.77	0.97	The closer score to 1 means its more stable	0.777

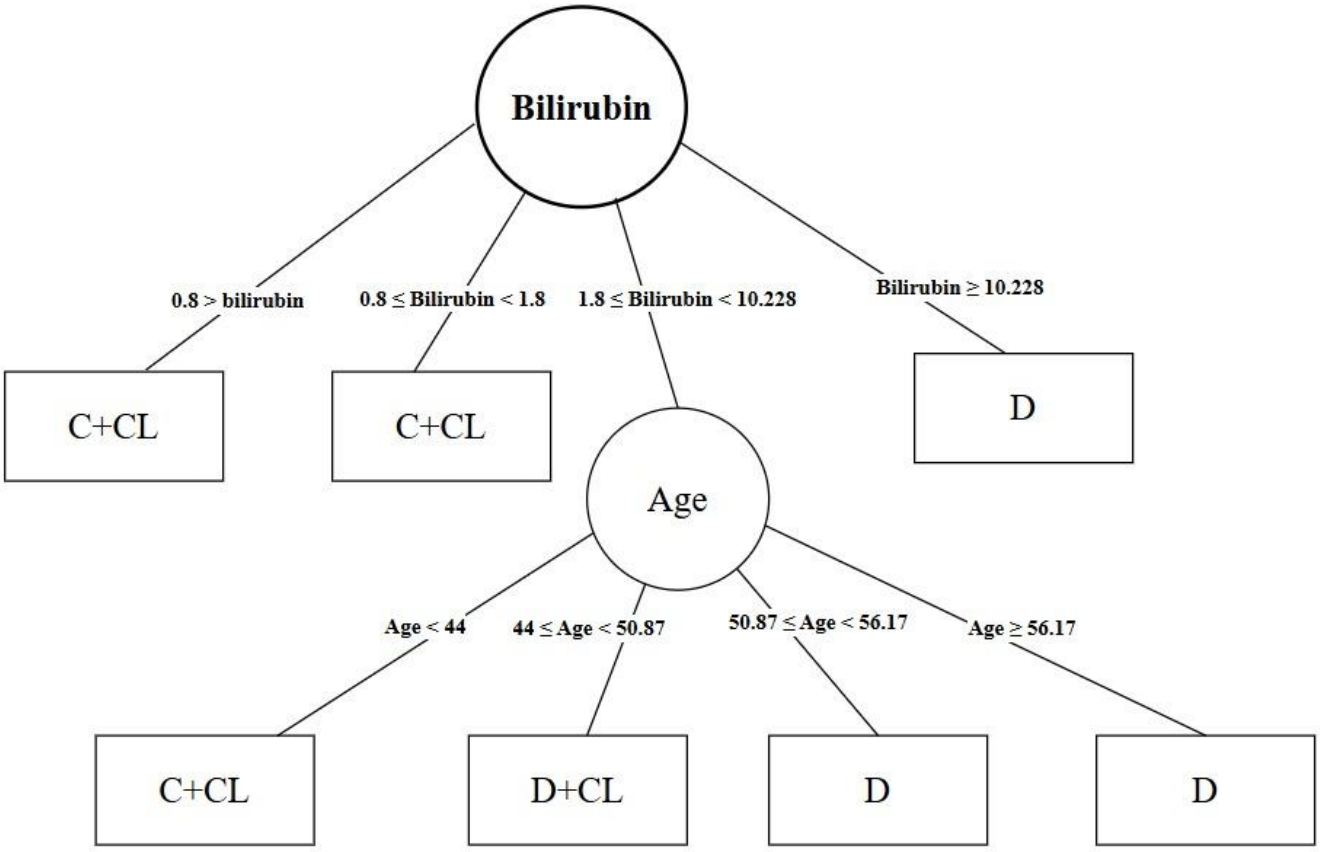
Overall Score = 0.6 * **Accuracy** + 0.05 * **Simplicity** + 0.2 * **Lift** + 0.15 * **Stability**

The 'Best' model is the **Decision Tree 12,** its characteristics:
Gini, with **maximum 4 branches**, **minimum leaf size is 18.**

Q. Try to find a combination different from the above combinations that gives the best results.

We tried 7 different combinations to find a combination that gives the best results, but none was better than what was already found.

1.3 Best Decision Tree Diagram:



Task 2: K-means Clustering

2.1 Segmentation:

s	Segmentation Label	Number of Clusters	Cluster Sizes	Size of Smallest Cluster	Outlier (Yes or No)
1.	km_2	2	146, 272	146	No
2.	km_3	3	136, 107, 175	107	No
3.	km_4	4	96, 80, 136, 106	80	No
4.	km_5	5	125, 64, 104, 41, 84	41	Yes

Q. *If the domain expert was interested in evaluating outlier clusters, which segmentations (i.e., sets of clusters) would you provide to the expert for evaluation?*

If the domain expert is interested in evaluating outlier clusters, we would provide the **km_5** segmentation for evaluation.

The dataset has **418 observations**, so 10% of the data is:

418 × 0.10 = 41.8, it means that any cluster with fewer than 41.8 observations can be considered an outlier cluster. Based on the table above, only **km_5** has an outlier cluster because one of its clusters contains 41 observations, which is less than **41.8**.

2.2 Descriptive Statistics:

The descriptive statistics show the range of the numeric variables in the cirrhosis dataset. The average **Age** of the patients is about **50.74**, with a **Standard deviation** of **10.45**, indicating age variation. Albumin has a 3.49 mean and a STD of 0.424, which means patients Albumin values are centered around the mean, other numeric values descriptive statistics are provided in the table.

Descriptive Statistics		
Variable	Mean	Standard Deviation
Age	50.741551	10.447214
Albumin	3.497440	0.424972
Alk_Phos	1800.851914	1876.389188
Bilirubin	3.220813	4.407506
Cholesterol	316.633971	112.021977
Copper	91.367225	75.095900
N_Days	1917.782297	1104.672992
Platelets	257.127990	97.093806
Prothrombin	10.730981	1.020029
SGOT	120.011555	49.536255
Tryglicerides	119.318182	54.308663

2.3 KM_3 Decision Trees

Cluster KM_3	Accuracy		Simplicity		**Overall Score**
	% Error	**Score**	**No. of Leaves**	**Score**	
Decision Tree 1	0.253	0.7470	6	1	0.8482
Decision Tree 2	0.4217	0.5783	2	0	0.3470
Decision Tree 3	0.4337	0.5663	5	1	0.7398
Decision Tree 4	0.3012	0.6988	7	1	0.8193
Decision Tree 5	0.4217	0.5783	1	0	0.3470
Decision Tree 6	0.2651	0.7349	8	1	0.8409
Decision Tree 7	0.3976	0.6024	8	1	0.7614
Decision Tree 8	0.3373	0.6627	7	1	0.7976

Decision Tree1 is the 'best' Decision Tree model in predicting **ClusterID** in km_3.

2.4 Explanation of the 'Best' DT results

Variable	**Cluster_1**	**Cluster_2**	**Cluster_3**	**Explanation**
Alk_Phos	x	x	x	Most important variable for predicting clusters
Age			x	Important variable predicting cluster 3
Bilirubin	x	x		Important variable predicting each of clusters 1,2
Narrative on comparison: ALK_Phos is the most important variable in the DT in predicting all clusters. After Alk_Phos, Age is an important variable in predicting most of cluster 3, and bilirubin is one of the most important variables in predicting most of the variables 1,2				

Task 3: Using KM_3 Biggest Cluster

Model	Accuracy		Simplicity		Overall Score
	% Error	Score	No. of Leaves	Score	
CHAID	0.2571	0.7429	1	0	0.4457
Gini	0.2571	0.7429	1	0	0.4457
Entropy	0.2571	0.7429	1	0	0.4457
IGR	0.2571	0.7429	1	0	0.4457

According to the table above, all models provided the same **Accuracy**, **Simplicity**, and **Overall score**. **Entropy** was marked as the champion model, So Entropy is the 'best' model for Task 3. Number of leaves is 1, might be due to the fact that the minimum leaf size is 30% of the ClusterID_3.
175 * 30% = 52.5 ≈ 53 minimum leaf size.

Q*. Compare the accuracy of the DTs of the cluster versus that of the 'best' DT of task2?*

For Task 2, our 'Best' DT Accuracy achieved **0.7470**, while Task 3 achieved **0.7429** for the 'Best' DT Accuracy. No big differences, so both models performed almost the same for Accuracy.

Examine hypotheses

The cirrhosis stage wasn't the best status predictor the most important status predictor was bilirubin. The higher cirrhosis stage is associated with a higher risk of death. The model successfully distinguished between D and C status, but failed to distinguish CL. This might be due to the class imbalance, where CL is the minority.